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Fröhlich's Coherent Excitations & The Cancer Problem—A Retrospective Overview of His Guiding Philosophy¹

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The guiding philosophy underlying Fröhlich's approach to biology from the side of theoretical physics is summarized, and illustrated, in the context of his prediction of (dynamic) coherent excitations in living systems, based on their dielectric and elastic properties, and far-from-equilibrium (nonlinear) character. His envisaged role of these coherent excitations in cell division and its control is outlined, together with the associated implications for cancer—as understood both at the time of his work and subsequently.

Keywords Fröhlich; Biological coherence; Cell division control; Cancer.

Introduction

In 1987, an international conference on the “*Biophysical Aspects of Cancer*” was held in Prague—the first of many to be organized there by J. Pokorný. One of the speakers at this first meeting was the internationally renowned theoretical physicist Herbert Fröhlich, who gave a paper entitled “Coherence in Biology.” The paper concluded with some speculative remarks on the possible role that his so-called “coherent excitations” might play in the control of cell division—a control that is absent in the case of cancer, from which he himself was to die in January 1991, within 4 years of the Prague conference.

The 21st anniversary of this historic conference is an appropriate occasion to recall and reflect² upon the basic philosophy underlying Fröhlich's approach to biology

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¹This article was partly based on an address given in the Prague *Carolinum* to mark the 660th anniversary of the foundation of Charles University, and the opening of the international symposium *Biophysical Aspects of Cancer: Electromagnetic Mechanisms*, July 1, 2008.

²To facilitate this, advantage is taken of the benefit of hindsight to integrate the themes developed in his many publications over 20 years into a coherent presentation of his philosophy and approach. Incidental references will be found in the footnotes, while those in the main body of the text are to papers in which key developments were first published.

from the side of theoretical physics, which dates from 1967, since this transcends³ the validity of the specific models of active biosystems that he later constructed to show *how* his ideas might be realized in practice.

One might reasonably ask: “What can theoretical physics possibly have to offer biology?” For biological systems are not only structurally far more complicated than those usually considered in physics, but they also have *evolved* to possess quite remarkable physical properties that enable them to function (when alive, or active) in highly precise and orderly ways, somewhat akin to machines.⁴

Now, while theoretical physics can never hope to predict the way in which this evolution has actually occurred,⁵ Fröhlich believed that theoretical physics could possibly be able to shed some light on the mystery of the evolved *bio-functionality*. Examples of such evolved properties of living systems include their dielectric behavior and their frequency-dependent hypersensitivity to ultra-weak external electromagnetic fields. In some cases, this sensitivity is limited only by quantum mechanics: the eye, for example, can detect a single light quantum, despite the fact that it is made from materials that no engineer would dream of using in the construction of an image converter!⁶

Although by 1967 there had been considerable progress in molecular biology in elucidating the structure of DNA, biomolecules and cells, this was not matched by a corresponding understanding of the various *activities* of living systems—apart from the fact that their orderly functioning was surely indicative of the presence therein of some kind of motional (or dynamic) *order*. “Order,” however, is a concept that *is* familiar to physics—wherein it pertains to the existence of some kind of macroscopic regularity in the physical behavior of the system as a whole; accordingly, it might be hoped that it is in this connection that an approach from theoretical physics could be fruitful. In general, physical order is brought about by *coherent* cooperation between the individual units that constitute the whole system, whereby the system becomes greater than the sum of its parts, in the sense that the emergent order is of a truly *collective* kind that is simply not meaningful on the level of the individual components.⁷

In searching for an appropriate physical way to describe functional order in biology, it is instructive to first consider the case of a machine, which no matter how complex its construction, can function in a relatively uncomplicated way that is describable in terms relatively few macroscopic variables,⁸ such as its center of mass, for example. Now the structure of a machine is the *same* whether or not it is functioning.

³He repeatedly stressed (e.g., Fröhlich, 1986, p. 243) that experimental verification of a biophysical prediction need not necessarily be considered as proof of the particular theoretical model used to make the prediction; other models may well lead to the *same* prediction. Such a multi-causal situation (which arises in consequence of the far-from-equilibrium, nonlinear character of living systems) necessitates very close collaboration between theory and experiment—much closer than is usual in physics. Indeed, assuming experiment confirms a particular prediction, it is possible, in the present case, to ask a question that is strictly forbidden in physics (except when dealing with machines), namely: What is the biological *purpose* of the predicted property?

⁴Fröhlich, 1986, p. 243.

⁵Fröhlich, 1980, p. 87.

⁶Fröhlich, 1980, p. 88.

⁷Just how large a system must be for collective behavior to be possible is governed by the magnitude of the fluctuations in the relevant macroscopic quantities (Fröhlich, 1970a).

⁸Fröhlich, 1970a, 1981.

In addition, the (microscopic) thermal motion of its constituent atoms is only minutely perturbed in the presence of the overlying, superimposed macroscopic dynamical order that characterizes its functioning; accordingly, the majority of its degrees of freedom remain unaffected by its functioning. These considerations hold also in the case of a biological system, a crucial discriminant of whose aliveness is its ability to *function* in an orderly way. The task of theoretical physics here, therefore, is to tentatively make suggestions as to what macroscopic order parameters of a physical kind might possibly underpin biofunctionality.

In attempting to pursue this, one is immediately faced with the difficulty that, with respect to their functioning, living systems are held far from thermal equilibrium by their metabolic activity. Despite this, they function in remarkably stable ways, although bio-functionality can, and does, alter depending on the state of a system's development; for example, an organ grows by cell division until it reaches its correct size when it stops, cell division then occurring only to replace damaged/dead cells—in such a way, however, that the original architecture is precisely restored. A metabolically active biosystem is an example of what in physics is called a *nonlinear* system—a system whose properties depend on the state in which the system (in the course of its development) finds itself,⁹ i.e., on its state of excitation. Accordingly, one cannot hope to be able to understand the dynamics underlying the functioning of a living system in any systematic way, starting from its equilibrium properties;¹⁰ instead, one must necessarily proceed non-systematically, essentially by way of inspired guesses.

In searching for an evolved physical property that is inherently allied to metabolism, the quite remarkable dielectric properties of living systems had suggested themselves to Fröhlich already in the late 1930s, when he learned¹¹ that across the dipolar plasma membrane of a (living) cell there is an enormous electric field¹² whose strength is of the order of 10 million V/m —larger than that used in the electric chair! Being an expert in dielectrics, he knew that in a field of this magnitude inanimate matter would undergo dielectric breakdown (unless special precautions were taken) thereby ceasing to behave as an electrical insulator. The fact that this evidently does not happen in the case of the material constituting the plasma membrane of a living cell suggested to him that this must be so for some good reason, probably connected with bio-functionality: for the transmembrane electric field vanishes in the absence of metabolic activity. Embedded in the membrane are large proteins that effectively partition it into *dipolar* units, the proteins themselves becoming polarized¹³ in the strong electric field. Already at this early stage, he went on to estimate the frequency at which the inner and outer surfaces of the membrane vibrate against one another, obtaining a value in the vicinity of 10^{11} Hz (100 GHz)—a frequency that, at that time, was not experimentally available.

⁹Fröhlich, 1970a.

¹⁰Fröhlich, 1980, p. 87.

¹¹Fröhlich, 1988, p. 1; see also Hyland, 1987.

¹²This field derives from the fact that because of metabolic pumping of Na and K ions the inner and outer surfaces of the cell membrane become oppositely charged electrically.

¹³By taking into account the shape dependence of the electric polarization (free) energy, he showed, through elaborations of some qualitative considerations dating from 1967 (Fröhlich, 1969), that the induced dipole moments can get (meta) stabilized as a (coherent) *quasi-ferroelectric* state (Fröhlich, 1973); for an earlier application of this to understanding enzyme action, see Fröhlich, 1970b.

Some 30 years later, however, experiments in the former U.S.S.R. revealed (Devyatkov, 1974) that living systems are actually highly sensitive to coherent, ultra-weak external electromagnetic radiation in this same frequency region, i.e., to mm microwave radiation—a fact that Fröhlich found out about only by accident in 1975 (Hyland, 1987). This was eight years *after* he had predicted (Fröhlich, 1968a) that, provided a minimum rate of metabolic energy supply is available, a steady state exists in which the lowest frequency longitudinal collective mode of electric polarization (based on the vibrations of all dipolar units of a given kind¹⁴ within a biosystem) becomes strongly excited¹⁵—far in excess of the value corresponding to thermal equilibrium. This arises due to a nonlinear interaction between the polarization modes and their surrounding medium (which is treated as a heat bath in thermal equilibrium) whereby, through energy exchanges involving *two* polarization quanta, the metabolic energy that is fed into the polar modes gets redistributed among them, eventually becoming channeled into the one with the lowest frequency, which accordingly becomes strongly excited. This *single*, supra-thermally excited mode, which then dominates the dynamical behavior of the system (whose other modes of vibration remain excited only to thermal levels, as they are in the absence of metabolic activity), is characterised by a single, sharp frequency. This macroscopically significant mode represents a highly ordered *coherent* electrical vibration of the whole system to which each dipolar unit contributes with the appropriate amplitude and phase; Fröhlich called this ordered vibration a “coherent excitation” (Fröhlich, 1968a).

In its simplest and most dramatic form, the coherent excitation takes the form of a so-called “giant dipole” vibration. In this (collective) mode of vibration the polarization is uniform (homogeneous)¹⁶ throughout the system, the positive charge of each dipolar unit moving in synchrony in the same direction by the same amount, and the negative charges similarly so in the opposite direction, so that the vibration of the whole system is effectively an N -fold magnification of any one of the N constituent dipolar units, i.e., it vibrates as a *single* “giant” dipole. There are three such modes whose frequencies are intermediate between those of the usual (bulk) longitudinal and transverse modes, and they depend (because of the long range of the Coulomb interaction) on the macroscopic size and shape of the system. These modes acquire particular significance in the case of systems whose spatial extent is less than the vacuum wavelength of electromagnetic radiation of the same frequency (e.g., as in the case of the interaction of mm microwave radiation with a biological cell) for which the distinction between longitudinal and transverse vibrational modes becomes meaningless (Fröhlich, 1949).

Replacing the energy supply from metabolism with that from sunlight, he later went on to adapt the model to cover *photosynthesis* (Fröhlich, 1968b).

Thus, instead of simply heating the system, and thereby increasing the amount of disorder therein, the influx of metabolic energy in conjunction with nonlinear processes results in the establishment of a counter-intuitive state of vibrational electric order, which is stabilised through other nonlinear effects.

¹⁴Not only those in the cell membrane and the other biological components considered originally (Fröhlich, 1968), but also in the subsequently discovered microtubules of the cytoskeleton (Fröhlich, 1986, p. 251).

¹⁵There is an evident parallel here with the case of the laser, which lases only above a certain threshold of optical pumping.

¹⁶For example, a mode of uniform polarization has infinite wavelength, or, equivalently, zero wave-vector.

This stabilization is based on another characteristic material property of a biological cell, namely, its elastic deformability (Fröhlich, 1969). The system exploits this property to lower its energy by changing¹⁷ its shape in an appropriate way, since the polarisation energy of the coherent electric vibration depends on the shape of the system. Essentially, by changing its shape the system creates an environment in which the coherent electric vibration sits “more comfortably,” whereby its energy is lowered and the vibration stabilized; the coherent excitation is then characterized by a threshold rate of energy supply that is lower than it would otherwise be, and by a frequency that is also lower (i.e., the mode is “softened” to some extent). This stabilization rests on the fact that, because of its inertia, the time required for the elastic deformation to occur is very much longer than the period of the coherent vibration. Such deformation is, however, resisted by the elastic restoring forces that determine the shape of the undeformed system, resulting in a macroscopically significant (coherent) quasi-static stress developing at the cell surface. Already in 1967, Fröhlich was quick to note the possible role this coherent surface stress might play in cell division; for with increasing size, the surface stress increases, which could thus act as a stimulus for cell division (Fröhlich, 1969); eventually, however, with progressive division, the cells will become so densely packed that the elastic deformation necessary for the stabilization of a cellular coherent mode cannot occur, whence the mode collapses, and with it vanishes the stimulus for any further cell division and growth, in accordance with the well-known phenomenon of contact inhibition.

Once he had learned of the Russian discovery that living systems are non-thermally sensitive to specific frequencies of very weak external mm microwave radiation, Fröhlich immediately published a *Physics Letter* (Fröhlich, 1975) claiming that this constituted experimental support for his coherent excitation. For, since there is no evolutionary need for living organisms to be sensitive to this kind of radiation, the fact that they *are* strongly suggests that they themselves make organised use of vibrations in the *same* frequency range (in the form of a coherent excitation) for their own internal purposes—such as, perhaps, in an electromagnetically based autonomous control system, *via*, which, for example, the vast number of biochemical reactions ($\sim 100,000$) that occur per second in a single cell are executed and orchestrated with the necessary speed and precision; for at 10^{11} Hz, even a quite narrow bandwidth can carry an enormous amount of information.

To resume then, we see that Fröhlich’s proposal to attack the problem of bio-functionality from the side of theoretical physics was based on the following three characteristics of living systems, which he first presented in 1967 (Fröhlich, 1969) at an international conference in Versailles organized by l’Institut *de la Vie*.

1. They possess some kind of motional order allied to their functioning.
2. They are relatively stable, and yet, with respect to their functioning, are far from thermal equilibrium.
3. The constituent cells and giant molecules possess notable material properties—namely their remarkable dielectric behavior and their elastic deformability.

Between his initial realization of the remarkable dielectric properties of living cells and his prediction therein of coherent excitations some 30 years later, Fröhlich continued his work in dielectric theory, out of which evolved such concepts as the polaron and so-called

¹⁷This depends on another kind of non-linear interaction involving a coupling between the polarization field and the elastic modes of the system (Fröhlich, 1973).

“hot” electrons. His most influential contribution (Fröhlich, 1952), however, was the introduction into condensed matter physics of the concepts of quantum field theory—with which he was already familiar from his work on the meson theory of nuclear forces during the late 1930s (Fröhlich et al., 1938). From this evolved what proved to be the essential ingredient to the eventual solution of the problem of superconductivity, namely, the possibility of an attractive interaction between electrons near the Fermi surface of a metal. This interaction, which is essential for Cooper pairing, is mediated by the exchange of lattice vibrational quanta (phonons), just as the nuclear force is mediated by the exchange of (virtual) mesons. It was from superconductivity, together with his subsequent work on other macroscopic quantum systems characterised by dynamic (or motional) order, that Fröhlich was led to appreciate the fundamental importance of the concept of *coherence*, and to consider the possibility that it might have a wider relevance—in particular, to biology (Fröhlich, 1969). Within the framework of quantum field theory, the highly correlated motion of distant parts of a coherent system expresses itself in terms of long-range correlations between the phases of so-called macroscopic wave-functions.

In 1972, five years after his prediction of coherent excitations in a living system, Fröhlich went on to show (Fröhlich, 1972) that two electrically neutral systems (such as cells) can *attract* one another over a distance that is much larger than the range of the familiar van der Waals’ force, provided they are properly tuned, i.e., oscillate coherently with the same frequency. If it is assumed that the vibrations are predominantly dipolar, then the new force is proportional to the inverse cube of the cells’ separation. Supporting evidence for such a long-range, frequency-specific attractive force, which depends on metabolic energy supply, an intact membrane and a transmembrane electric field, came from studies of rouleaux formation in red blood cells (Rowlands, 1983; see also Rowlands, 1994, wherein references to earlier work can be found) in which, below a certain separation, the cells rush together to form coin-like stacks (*rouleaux*) at a much faster rate than is expected on the basis of Brownian motion. These results indicated that the force operates over a distance of the order of $10\ \mu\text{m}$ (i.e., 10 times the typical range of chemical forces), and that in a mixture of blood cells from different organs/mammals, rouleaux formation occurs predominantly between cells of the *same* species, consistent with them “recognizing” one another through their common frequency. It should be appreciated that this new frequency-selective interaction can facilitate a whole sequence of biochemical events; for provided a minimum rate of energy supply is available to establish the cellular coherent oscillations in the first place, a force can come into play, which is dependent (through the associated electro-strictive deformation) on the state of development of the system, and which, in turn, can influence its future development.

Fröhlich’s coherent excitation, in which a single collective electric vibrational mode is excited to a larger-than-thermal amplitude, is, quantum mechanically speaking, equivalent to a macroscopically large number of polarisation quanta all having the same frequency. The sought-after macroscopically significant (order) parameter (in terms of which the activity of a living system might be expressed) is thus seen to be allied to the macroscopic occupation of a *single* quantum state, i.e., the coherent excitation is a kind of Bose condensate¹⁸ (Fröhlich, 1968a): in this

¹⁸The possibility of Bose-like condensation here arises from the competition between the dissipation inherent in the system and the metabolic pumping, which effectively *fixes* the total number of polarization quanta; this permits a non-zero chemical potential to be (formally) defined, and the attendant possibility of Bose-Einstein condensation, as in a gas of a fixed number of material particles.

respect, a living system must be considered as an *open, non-equilibrium, room temperature macroscopic quantum system*. As such, living systems must be anticipated to be sensitive to electromagnetic vector potential fields (Smith, 1998), and there is some experimental evidence¹⁹ that this is indeed the case. It should be noted that such sensitivity would endow living systems with the possibility of a quite novel form of communication that would escape conventional detection.

There is now a variety of experimental support in the case of mono-cellular organisms for Fröhlich's coherent excitation—but not for its realization as a simple giant dipole. As has been pointed out by Pokorný (2009), it now seems likely that the excitation is based on the strong electric dipoles in microtubules of the cytoskeleton, as earlier²⁰ envisaged by Fröhlich, which are pumped by energy (surplus to that utilised for the production of ATP) that is produced by the mitochondria, and which is usually considered to be waste energy. The microtubule frequencies range from kHz to GHz (Pokorný, 2009). In the kHz range, coherent mechanical vibrations (at around 1 kHz) of the cell membrane (with which the microtubules connect, and which are allied to metabolic activity) have been reported (Pelling et al., 2004); attempts to replicate this finding are currently in progress at the Czech Academy of Science in Prague. In addition, short-range surface electric fields have been detected in the kHz–MHz range, around a variety of cells, particularly near mitosis,²¹ using a variety of techniques (Pohl and Pollock, 1987; Pollock and Pohl, 1988; Smith et al., 1987). At GHz frequencies, on the other hand, non-thermal influences on the growth rate of yeast (Grundler et al., 1977; Grundler and Kaiser, 1992) and certain bacteria²² (Fröhlich, 1980) have been found.²³

It must be said, however, that the biological “purpose” of such coherent excitations remains obscure; yet purpose must be established²⁴ before it can be claimed²⁵ that these excitations play a fundamental role in any new theory of biofunctionality. In this connection, it should be stressed that a coherent excitation is not necessarily a permanent feature of a living system, but may be “switched-on” only if and when required in connection with the execution of some vital function.

¹⁹Ho and Saunders, 1994; Kashulin and Roldugin, 1999.

²⁰Fröhlich, 1986, p. 251.

²¹Prior to mitosis, there is also an increase in biophoton emission in the *visible* range (Popp et al., 1981).

²²Other experiments utilising synchronised bacterial cells have detected shifts in the laser-Raman spectra of metabolically active *B. Megaterium* & *E. Coli*, corresponding to resonances between 7×10^{10} and 5×10^{12} Hz, which appear, however, only at certain stages of evolution (Webb and Stoneham, 1977). In addition, an anti-Stokes/Stokes ratio close to unity (instead of the thermal value of about 0.5 at physiological temperature) has been measured in *E. coli*, consistent with a strongly supra-thermally excited mode in the system (Webb et al., 1977). Details of these experiments (and others) can be found in Fröhlich's review article of 1980 (Fröhlich, 1980).

²³It should be noted that the failure of other attempts (e.g., Furia et al., 1986; Gos et al., 1997) to reproduce the effects found in yeast are most likely due to crucial *differences* in experimental protocol—such as irradiating at different stages in the cell cycle, differences in the way in which growth rate is measured, differences in cell density, differences in synchronization methods, etc.—which effectively undermine the fidelity of the purported replications.

²⁴The possibility (and indeed the necessity) of establishing “purpose” distinguishes biology from physics, where such enquiries are permissible only when dealing with machines (Fröhlich, 1983, p. 1; 1986, p. 243).

²⁵Fröhlich, 1988, p. 21.

Turning now to the control of cell division in multi-cellular systems, it was clear to Fröhlich that some kind of collective behavior must here be involved, since the rate of division is governed by the size and shape that an organ has reached at each stage of growth, division ceasing once the organ is fully grown—except when it is required to replace damaged/dead cells: such control is absent in the case of cancer. His idea (Fröhlich, 1977a, 1978), dating from the mid 1970s, was that the required control is perhaps based on a global coherent electric vibration that extends throughout an entire organ, to which the coherent vibrations of each individual cell (which he assumed was based in the DNA protein complex in the cell nucleus) contribute with the appropriate phase,²⁶ the frequency being a characteristic of the particular organ in question. Because of the long range of the interactions involved, such a global excitation itself depends on the size and shape of the system,²⁷ and its spatial structure can, in general, be expected to be quite complicated and only very weakly optically active;²⁸ in addition, the long-range attraction between cells oscillating coherently with the same frequency contributes to the stability of the tissue.

Owing to their participation in the global coherent excitation of the whole organ, individual cells lose their autonomy, resisting, in particular, any attempt to have their own coherent frequency altered. Should a change in a cell's coherent frequency be a necessary precursor of cell division (as seems likely in view of the necessity of DNA to first uncoil), division will be inhibited unless energy is available to liberate the cell from the global coherent mode in which it is frequency/phase locked; the energy available for this purpose (and also that to pump the global coherent excitation) comes from nutrients, the amount of which gets progressively depleted. With successive cell division, the strength of the global vibration is increased with the participation of the daughter cells (which are frequency-entrained back into resonance²⁹), and it thereby exerts a stronger influence over the autonomy of individual cells; accordingly, more energy (*less* of which is now available) is then required to free a cell so that it can divide. These two negative feedback effects re-enforce one another, so that cell division becomes a *self-limiting* process, eventually ceasing all together (except when necessary to replace damaged/dead cells), when the organ stops growing.

Within this scenario, Fröhlich's approach to cancer was the following. If, for some reason—such as an accumulation of intra-cellular mutations, or because of invasion by foreign molecules (carcinogens)—some cells lose the “correct” frequency, they will

²⁶The global coherent vibration can be excited only once the organ has reached a size such that the number of contributing cells is large enough to permit the necessary *collective* polarization mode to be defined—see Footnote 7. Below this size, cells can divide freely, the electrostrictive stress at the cell surface mentioned above acting as a stimulus. Once the cells are locked into the global coherent excitation, however, this stimulus ceases, since the deformations of individual become sublimated into an overall deformation of the whole organ; this affords an alternative basis (to that given above) for the so-called “contact inhibition” exhibited by cells in normal tissue.

²⁷The frequency of the global coherent excitation must be expected to differ somewhat (because of the nonlinear electrostrictive coupling referred to above) from that of an individual cell in isolation.

²⁸Since (unlike the case of a cell) the size of an organ can be much *larger* than typical electromagnetic wavelengths, it is possible that the global coherent excitation is here predominantly *longitudinal*.

²⁹After division, once the DNA complex has recoiled, the frequency will revert to the value corresponding to a cell in isolation, which, in general, differs from that of the global coherent mode; see Footnote 27.

thereby cease to be in resonance with the global excitation. Provided not too many cells are so affected, the global coherent vibration will be only slightly weakened, and still strong enough to exert control over the division of the remaining untransformed cells, and to entrain newly divided cells back into resonance (provided their frequency is not too different from that of the global coherent excitation). If, however, the number of aberrant cells exceeds a certain critical value, the global coherent excitation will be weakened to such an extent as to be incapable of exerting these controlling and corrective influences. Thus, there occurs a kind of phase transition to a disordered state of uncontrolled cell division characterized by a loss of coherence: this state, Fröhlich identified with *cancer*.

In this connection, it is of interest to note that a report in *Nature* of a meeting (*Oncogenes and Growth Control*) held ten years later at the European Molecular Biology Laboratory in Heidelberg contained the following statement: "Perhaps the most interesting new theme of the meeting was that normal cells, or their products, can exert control over those of their number that have been transformed into cancer cells." (Newmark, 1987).

At the end of his 1977 review article (Fröhlich, 1977a), Fröhlich speculated whether it might be possible to restore control of cell division to this disordered (coherenceless) state by exposing the system to a sufficiently strong external electromagnetic field whose frequency matches that of the endogenous global coherent electric vibration that has here broken down, concluding: "This, of course, would be the most startling influence of electromagnetic waves on biological systems." Many years later, this was to some extent realised in the context of so-called "microwave resonance therapy," developed in the Ukraine during the 1990s (Sit'ko et al., 1996).

Although the mechanism differs from that envisaged by Fröhlich, it is of interest, in connection with the apparent ability of an external oscillatory electric field of sub-thermal intensity to exert a corrective influence on cancer, to note a recent advance to which researchers in Prague have contributed (Kirson, Dbaly et al, 2007). Rather than re-establishing control of cell division by re-instating a controlling coherent vibration, the applied electric field in the kHz region (the precise frequency of which is organ-dependent) here appears to act by delaying cell division and by inducing apoptosis of dividing cancerous cells, leaving normal cells unaffected.

It is clear, however, that, within Fröhlich's cancer scenario, uncontrolled cell division occurs not only within the population of transformed cells, but also within the untransformed host tissue (hyperplasia). This is totally at variance with the *mono-clonality* of most spontaneous tumours, both malignant and benign, i.e., that the tumor originates from a *single* transformed cell, which then multiplies by uncontrolled division. The closest his model approximates to reality is when, due to external factors (such as exposure to ionising radiation) a *significant* number of normal cells are malignantly transformed (so called poly-clonality); in this case, however, cell division within the remaining normal tissue is still controlled, i.e., there is no accompanying hyperplasia. Despite these difficulties (which can be rectified as indicated below), they do not materially detract from the importance of his identification of the cancerous state as one in which the *coherence* that characterizes the non cancerous state is *absent*—an identification which later received experimental support from biophoton studies (Popp and Chang, 1998) in the visible region. These revealed that photon emission from cancerous tissue is more abundant and *much less coherent* than that from normal tissue; furthermore, the emission displays an opposite dependence on cell density, in parallel with the absence of contact inhibition among cancer cells.

The mono-clonality of tumours can be accommodated within the context of Fröhlich's ideas by assuming that in the course of its tumourigenic transformation the coherent frequency of the single progenitor cell is either: (a) lost altogether—as argued by Pokorný in the case of malignant tumours (Pokorný, 2008), or (b) changed to such an extent that it *cannot* be entrained back into resonance with the global coherent mode based on the remaining normal (i.e. untransformed) cells; in both cases, the aberrant cell remains free to divide and establish a population of clone cells within the normal host tissue—i.e. to form a tumour.

In case (a), there is clearly no possibility of a global coherent excitation, whereby cell division within the clone population might be controlled, simply because there are here no *cellular* coherent excitations to form the basis of such a global coherent mode.

In case (b), the problem is to understand how cell division can be controlled within the normal host tissue, but *not* within the clone population (assuming its cells individually support coherent excitations), and how differences between malignant and benign tumours might reflect themselves in this context. In terms of Fröhlich's ideas, this translates to understanding how a global coherent excitation strong enough to control cell division can exist in the population of (untransformed) normal cells, but *not* within the population of clone cells, be they malignant or benign. Two possibilities here present themselves:

1. A global coherent excitation is established, but is too weak to exert the necessary control; within the framework of Pokorný's scenario, this can be attributed to the partial disintegration of the cytoskeleton of malignantly transformed cells.
2. A global coherent excitation is *not* established, either because of an inadequate supply of energy,³⁰ or, in the case of malignant cells, because of de-differentiation, should this result in a *variety* of *different* cellular coherent frequencies, as seems reasonable.

De-differentiation is probably the most important qualitative difference between malignant and benign cells. In the benign case, where de-differentiation does not occur, there is no associated impediment to the possibility of a global coherent excitation, and its *inability* to exert the necessary influence to be able to control cell division must here be attributed to the mode being of *insufficient* amplitude. With progressive division, however, the global coherent mode will increase in amplitude as more and more daughter cells contribute to it, eventually becoming possibly strong enough to exert the required control, thereby arresting any further growth; consistent with this is the fact that the growth of benign tumours does indeed sometimes spontaneously arrest.

In both the malignant and benign cases, the energy required to pump any global coherent mode comes from that left over after the cellular coherent excitations have been established. At the present stage of development, however, it is not possible to quantify how this might differ even between normal and malignant cells, although the *threshold* pumping rate for the establishment of a *cellular* coherent excitation must be expected to be *lower* in the case of malignant cells, on account of their increased deformability (Guck et al., 2005; Suresh, 2007); within the framework of existing model calculations (Bhaumik et al., 1976), further quantitative comparison requires, however, also information on the density of normal tissue relative to that of a malignant tumour.

³⁰This depends on the amount of nutrient available, which is not unlimited, particularly in the case of glucose-addictive malignant cells.

In the benign case, where there the complication of de-differentiation does not arise, quantitative comparisons are again not possible because of the current absence of the information that *is* partly available in the malignant case concerning, namely, the cell's energy budget (level of mitochondrial dysfunction, degree of reliance on glycolysis, *etc.*) and the extent to which its cytoskeleton might be disorganised.

From the above considerations, it might be tempting to assume, as a first approximation, that malignant and benign tumours are associated with cases (a) and (b), respectively. This however, leads to difficulties in understanding certain distinguishing characteristics of the two types of tumour—difficulties that do not arise if *both* kinds of tumour are represented by case (b), the difference between them arising from the de-differentiation that characterises the malignant case. Thus:

- (i) *Benign tumours remain localised—i.e., are non-invasive and do not metastasise—unlike malignant ones:* This difference can be attributed, at least in part, to the frequency-specific long-range attractive force between benign tumour cells (each characterised by the *same* coherent frequency) being much stronger than between the cells of a malignant tumour, which favours localisation in the benign case. The weaker attraction between malignant cells—which facilitates metastasis—is due not only to the progressive loss of inter-cellular resonance arising from changes in cellular coherent frequency that most likely accompanies de-differentiation, but, within Pokorný's scenario, also to the reduction in the amplitude of the coherent mode entailed by the partial disintegration of the cytoskeleton (which might itself follow the degree of de-differentiation). A further contribution to the greater cohesion in the case of a benign tumour is the absence of any resonant attraction between its cells and those of the normal host tissue, on account of the *different* coherent frequencies that characterise the two kinds of cell. In the malignant case, on the other hand, because of the likely *variety* of cellular frequencies (entailed by de-differentiation), the possibility exists that some malignant cells are in resonance with those of the host tissue, whence the frequency-specific attractive force is operative, facilitating local invasion into the host.
- (ii) *Malignant tumour cells have much more irregular sizes and shapes than do those of benign tumours:* This is to be expected in view of the multitude of different electro-strictive surface stresses—and hence different cell shapes and sizes—associated with the range of cellular coherent frequencies that characterise the malignant case, to which any de-differentiation-dependent variation in cellular deformability will, of course, have contributed.

The above considerations focus attention not only on biophysical differences between malignant and normal cells, but also between the cells of benign and malignant tumours, about which there appears to be much less information available, particularly concerning differences in metabolism and the integrity of their cytoskeletons, both of which are intimately implicated in coherent excitations and the associated frequency-specific long-range inter-cellular attraction. It would also be of interest to investigate whether malignant and benign cells differs also in the values of their plasma and mitochondrial membrane potentials, and whether any differences as might exist correlate with differences in mutation number and level of mitochondrial dysfunction.

Conclusion

In conclusion, it can be said that the real importance of Fröhlich's potentially revolutionary contribution to biology³¹ is that it offers a novel theoretical framework within which a start can be made from the side of physics to understand the orderly functioning of living systems in terms of *coherence* based on their dielectric and elastic properties. In assessing the originality of his work, it must be remembered that it dates from almost 40 years ago when understanding of intra-cellular structure, nonlinear processes, and cancer genetics and microbiology was far less advanced than now. Nevertheless, his work continues to generate considerable interest, not only on account of the novel possibilities it offers for understanding cancer in the light of subsequent developments in this field—with the potential of suggesting novel therapeutic modalities—but also because of the variety of possibilities it offers for understanding the non thermal ultra-sensitivity of living systems to very weak electromagnetic radiation at specific frequencies,³² and certain aspects of brain function³³ and consciousness.³⁴

Nowhere in his long and distinguished career was Fröhlich's holistic outlook better illustrated than by his brilliantly daring suggestion that the concept of *coherence* might have relevance beyond the inanimate realms wherein it first arose—in particular, to **BIOLOGY**.

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³¹Good reviews, containing more technical aspects than considered here, can be found in Fröhlich, 1980, 1986, 1988, and Hyland, 2002.

³²For example, the ability of ultra-weak external microwave radiation of a specific frequency to “switch on” a particular bio-effect becomes understandable if the radiation supplies the *deficit* between the rate of energy supply available from metabolism and the threshold value required for the establishment of an associated coherent excitation (Fröhlich, 1980).

³³This is based on electrochemical *limit cycles* arising from a synthesis of the dynamic coherent excitation discussed above and the (static) one constituted by the quasi-ferroelectric state mentioned in Footnote 13 (Fröhlich, 1977b; see also Hyland, 2002).

³⁴See, for example: Globus et al., 2004.

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